

Project 2 — Authoring Kernels

1. How long did it take to simulate the color-gradient kernel?

The 256×256 run took 0.865 s end-to-end on the host. The simulator reported 1 s for its simulation-loop timer, rounded to whole seconds.

2. In what color format is color written to the framebuffer?

32-bit 0xAABBGGRR: red in bits 0–7, green in 8–15, blue in 16–23, and alpha in 24–31. The little-endian byte order is RGBA.

3. What is an execution model? What constitutes TRaX's execution model?

An execution model specifies how threads execute, are grouped, and interact. TRaX's model consists of threads and tiles: each thread executes on one TP, and each tile's threads execute within one TM. A default tile contains 32 consecutive work IDs.

4. What purpose does `fchthrd()` serve?

`fchthrd()` atomically obtains a unique work ID, allowing available execution contexts to claim more work through tile scheduling. Repeated calls let the kernel process more pixels than there are hardware contexts.

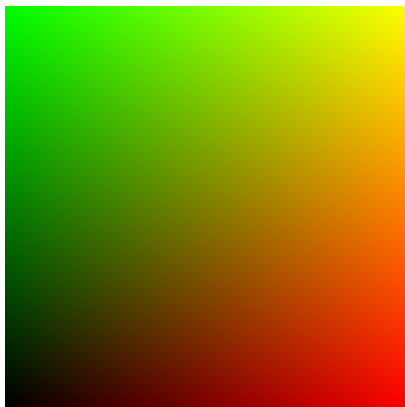


Figure 1: Color gradient, 256×256 .

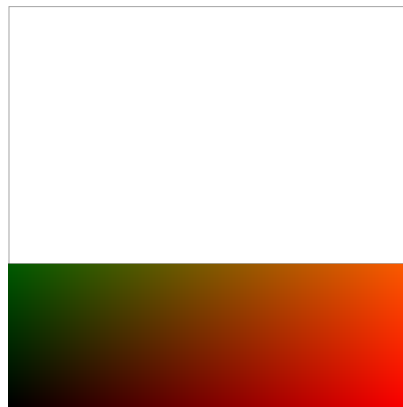


Figure 2: Without the loop, 256×256 .

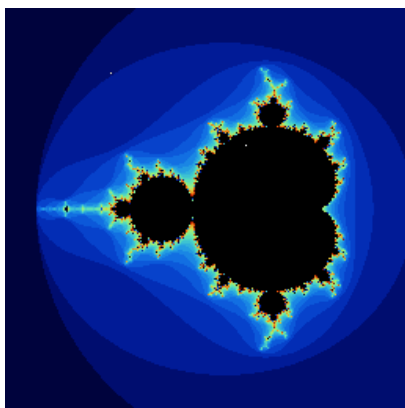


Figure 3: Mandelbrot trial, 256×256 .

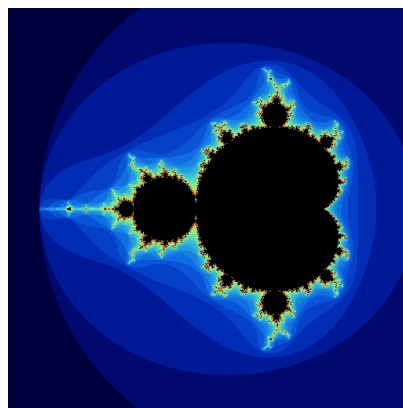


Figure 4: Mandelbrot submission, 512×512 .